Clean-Room Reconstruction and Physical-Consistency Audit of the Halekas et al. (2008) Lunar-Potential Example

ESPG Lunar Potential Map Project

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Abstract

We perform a four-phase clean-room reconstruction of the Halekas et al. (2008) Lunar Prospector Electron Reflectometer (LP ER) surface-potential example near 14:45 UT on 29 April 1999. A literal public reconstruction initially obtains $U_{\mathrm{rel}} = -261$ V rather than the published -171 V. Integrating a sharp physical boundary over actual angular support and applying the documented fractional energy response moves the joint estimate to -161 V under two normalization orders. Removing the beam-amplitude rail and fitting the measured broad beam gives a stable beam-only scale near -145 V and joint estimates of -151 to -161 V, but boundary-only estimates remain strongly dependent on cone refilling, broadening, masking, and magnetic ratio. A direct Figure 5 raster comparison then favors spectrum 654 over the previously assumed 653; its beam amplitude matches the publication, but its extracted loss-cone boundary does not independently select the published $(U_{\mathrm{rel}}, B_{\mathrm{S}}/B_{\mathrm{M}})$ pair. Publication-like boundary voltages appear when $B_{\mathrm{S}}/B_{\mathrm{M}} = 0.975$ is imposed, while free fits prefer substantially different field ratios. We conclude that the upward secondary beam robustly establishes a negative spacecraft-relative voltage scale of order 145–160 V in the benchmark interval. The public data and reconstructed method do not provide a stable independent two-parameter loss-cone confirmation of the exact published -171 V, 0.975 pair.

1 Purpose and Epistemic Status

The purpose of this experiment is not to tune a modern estimator until it matches a published number. It is to distinguish four questions:

1. Can the public description reproduce the historical result?
2. Which instrument and representation choices control reproduction?
3. Do the beam and loss-cone boundary independently support one voltage?
4. Does the archived Figure 5 distribution uniquely support the published voltage and magnetic ratio?

The published value is a historical reference, not ground truth. Stable disagreement may indicate a missing private processing detail, different manual quality control, or bias in the published point estimate.

The evidentiary basis is frozen in `provenance/halekas-cleanroom-reconstruction.md`, with a SHA-256 checksum. Phase A4 additionally freezes the native embedded Figure 5 data raster and its color-calibrated value map. Publicly unresolved choices are represented as explicit branches rather than silently selected defaults.

2 Data and Physical Observables

The LP ER three-dimensional product contains 15 sequential energy rows and 88 look directions. The benchmark neighborhood consists of spectra 652–654. The publication reports

$$U_{\mathrm{sc}} = +11 \text{ V}, \qquad U_{\mathrm{M}} = -160 \text{ V}, \qquad U_{\mathrm{rel}} = U_{\mathrm{M}} - U_{\mathrm{sc}} = -171 \text{ V}, \qquad B_{\mathrm{S}}/B_{\mathrm{M}} = 0.975. \tag{1}$$

The adiabatic boundary equation is

$$\sin^2 \alpha_c(E) = B_{\mathrm{S}}/B_{\mathrm{M}} \left(1 + \frac{U_{\mathrm{M}}}{E - U_{\mathrm{sc}}}\right), \tag{2}$$

where the reflected-side boundary plotted at high pitch is $\beta(E) = 180^\circ - \alpha_c(E)$. The upward secondary beam is centered at

$$E_{\mathrm{beam}} = U_{\mathrm{sc}} - U_{\mathrm{M}} = -U_{\mathrm{rel}} \tag{3}$$

in electron-volt/volt numerical units when the beam reaches the spacecraft.

The measured quantity is a normalized reflected-to-incident distribution. Two public-evidence normalizations remain plausible: division by a conjugate incident direction and division by the incident-hemisphere mean. The 2008 method minimizes an unweighted squared difference of logarithmic normalized fluxes over the complete distribution. This objective is descriptive; it has no recovered variance scale and its differences are not confidence levels.

3 Phase A1: Literal Public Reconstruction

Phase A1 rebins the 88 sectors into 16 fixed 11.25-degree pitch bins, uses a hard loss-cone boundary, tests cone floors 0.05 and 0.10, and represents the beam by one energy row and two field-aligned pitch bins. Beam amplitude is either fixed from the measured excess or profiled on a nuisance grid. Eight branches cross normalization, floor, and amplitude handling.

All branches select

$$U_{\mathrm{M}} = -250 \text{ V}, \qquad U_{\mathrm{rel}} = -261 \text{ V}, \qquad B_{\mathrm{S}}/B_{\mathrm{M}} = 0.910. \tag{4}$$

The result is stable but fails the predeclared 25 V historical voltage gate. The measured beam occupies both the 113 and 196 eV rows, while the synthetic beam is assigned to only one. The measured loss-cone transition is graded but the model is binary. Phase A1 therefore diagnoses a representation failure, not evidence for a physically exact −261 V potential.

4 Phase A2: Instrument Representation

Phase A2 freezes a $2 \times 2 \times 2$ comparison:

- bin flux then divide versus divide native sector pairs then bin;
- evaluate the sharp boundary at the pitch-bin center versus average the sharp sector-level model over the sectors that contributed to the bin;
- assign the beam to the nearest energy row versus apply a Gaussian response with $\mathrm{FWHM}/E = 0.48$.

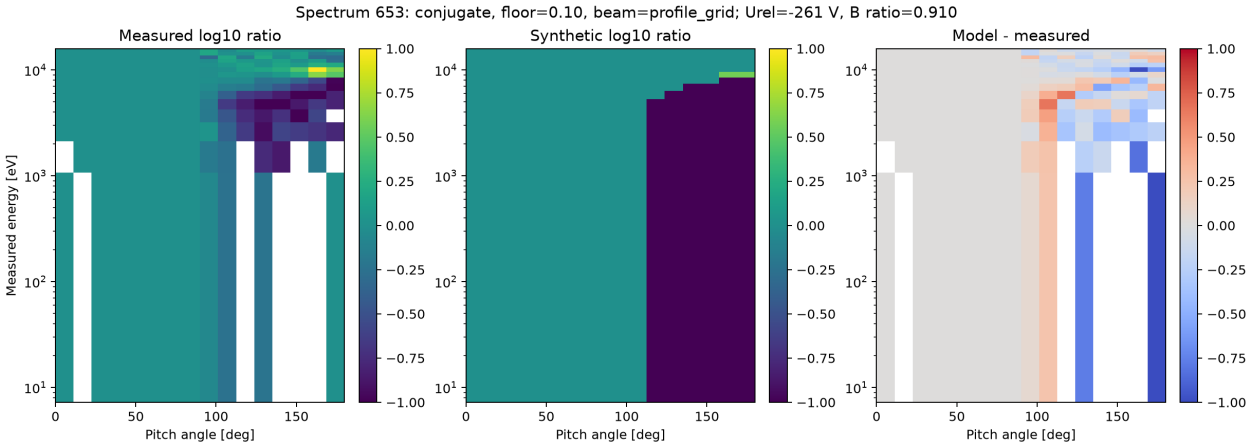


Figure 1: Phase A1 measured, synthetic, and residual distributions for the lowest-objective literal branch.

The cone floor is 0.10 and beam amplitude is profiled on $[0, 8]$.

Nearest-row branches remain at $U_{\mathrm{rel}} = -251$ to -261 V. The two support-integrated fractional-response branches agree exactly despite reversing normalization order:

$$U_{\mathrm{rel}} = -161 \text{ V}, \qquad B_{\mathrm{S}}/B_{\mathrm{M}} = 0.880, \tag{5}$$

only 10 V from the historical voltage. Thus the initial discrepancy is primarily a discretized beam-response artifact, with angular support integration required for representation stability. No arbitrary sigmoid is needed to recover the historical voltage scale.

However, both publication-like branches place beam amplitude at the frozen upper limit of 8. Independent fits give a beam voltage near -145.5 V and a boundary-only voltage near -281 V. Reproducing one joint scalar does not show that the two observables agree.

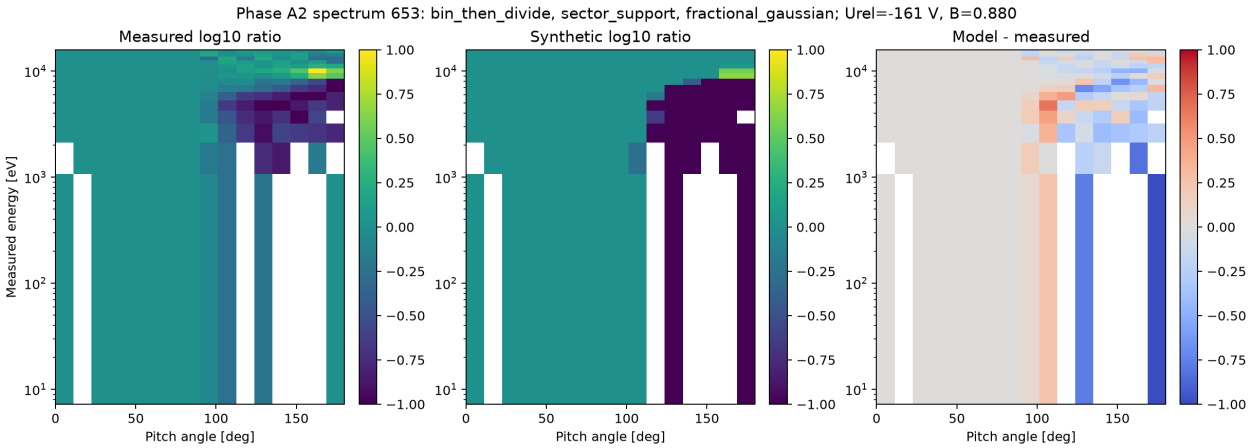


Figure 2: Best Phase A2 support-integrated, fractional-response comparison.

5 Phase A3: Physical-Consistency Stress Test

Phase A3 removes the artificial beam rail and crosses two normalization orders, narrow versus broad beam pitch response, sharp versus five-degree broadened boundary, and cone floors 0.10 versus 0.25. Fractional energy response is held fixed. Beam amplitude is profiled on $[0, 20]$.

All 16 joint branches return $U_{\mathrm{rel}} = -151$ or -161 V without amplitude or voltage rails. The lowest descriptive objective uses a broad Gaussian beam, five-degree boundary, and floor 0.25:

$$U_{\mathrm{rel}} = -151 \text{ V}, \qquad B_{\mathrm{S}}/B_{\mathrm{M}} = 1.000, \qquad A_b = 14.0. \tag{6}$$

The objective per nominal degree of freedom falls from 0.1938 in A2 to 0.1233, although structured high-energy residuals remain.

The independent two-dimensional beam fit is highly stable across normalization:

$$U_{\mathrm{relbeam}} = -145.4 \text{ V}, \qquad A_b \simeq 12.2, \qquad \mathrm{FWHM}_\alpha \simeq 58^\circ. \tag{7}$$

Boundary-only results, by contrast, span -281 V to $+9$ V as floor, broadening, and masking change. The joint estimate is beam-dominated. The loss cone does not provide a stable independent voltage under this model family.

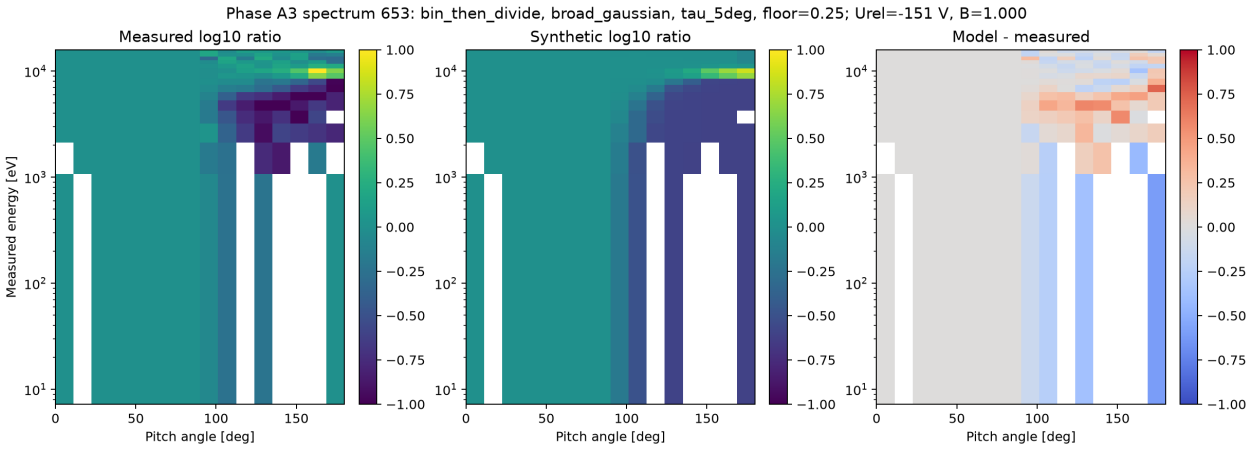


Figure 3: Best Phase A3 model after extending beam amplitude and angular support. Coherent residual structure remains despite the lower objective.

6 Phase A4: Figure Identity and Boundary-First Inference

Phase A4 stops fitting a joint beam-plus-cone voltage. It first compares spectra 652–654 with the frozen Figure 5 raster, then extracts a transition at each energy before applying equation (2).

6.1 Figure 5 identity

The primary morphology metric is reflected-side logarithmic RMSE with no fitted rescaling. Table 1 gives the closest normalization per spectrum.

Spectrum 654 begins at 14:45:28 UT and is a better panel candidate than spectrum 653 despite 653 having the closest midpoint to the rounded 14:45 label. With conjugate normalization, 654 has RMSE 0.417, correlation 0.932, and beam peak 4.51. Both representations favor 654. The match remains imperfect, so private processing or display differences are not excluded.

Table 1: Figure 5 panel-identity comparison.

Spectrum	Normalization	Log RMSE	Correlation	Beam peak
652	conjugate	0.646	0.873	15.28
653	incident mean	0.434	0.915	10.39
654	incident mean	**0.399**	0.926	**4.35**
Published	–	–	–	4.48

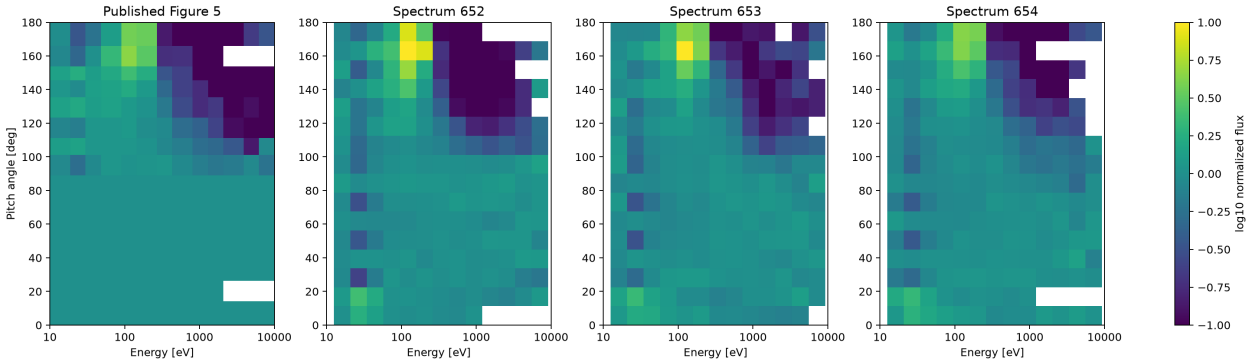


Figure 4: Frozen published Figure 5 raster and incident-mean candidates. Spectrum 654 most closely reproduces the beam amplitude and broad morphology.

6.2 Boundary extraction

For each native reflected sector at 300–10000 eV, Phase A4 fits

$$R(\alpha) = R_{\mathrm{in}} + \Delta R \left[1 + \exp\left(-\frac{\beta - \alpha}{w}\right)\right]^{-1}, \tag{8}$$

where the cone level, contrast, boundary β, and width are all free. One hundred deterministic sector bootstraps estimate boundary uncertainty, with a 5.625-degree resolution floor. The extracted boundary points are then fit with equation (2) under free $(U_{\mathrm{M}}, B_{\mathrm{S}}/B_{\mathrm{M}})$, fixed published $B_{\mathrm{S}}/B_{\mathrm{M}} = 0.975$, and a high-energy asymptotic field ratio.

The archived magnetic vector is exactly repeated across all 15 rows of each spectrum. Row-field and sweep-mean pitch calculations are identical to 10^{-13} degrees. Energy-correlated magnetic rotation is not responsible for the observed boundary behavior.

Under preregistered acceptance, spectrum 652 shows a coherent weak-potential trend and gives $U_{\mathrm{rel}} = -41$ to -61 V with $B_{\mathrm{S}}/B_{\mathrm{M}} = 0.975$–$0.990$. Spectrum 653 is non-monotonic and gives -111 to -151 V while requiring $B_{\mathrm{S}}/B_{\mathrm{M}} = 1.005$–$1.050$; conjugate fits reach the upper field-ratio edge. Spectrum 654, the closest panel, gives only -1 to -41 V when both parameters are free, with $B_{\mathrm{S}}/B_{\mathrm{M}} = 0.750$–$0.825$.

Several per-energy transitions hit nuisance bounds. The preregistered results remain primary, but a labeled strict check excludes those points. Spectrum 652 remains near -31 to -61 V; spectrum 653 loses the low-energy points carrying its negative voltage and moves near zero; spectrum 654 gives free fits around -61 to -81 V with $B_{\mathrm{S}}/B_{\mathrm{M}} = 0.755$–$0.795$. If $B_{\mathrm{S}}/B_{\mathrm{M}} = 0.975$ is imposed, strict spectrum-654 boundaries instead imply $U_{\mathrm{rel}} = -181$ to -191 V. Those fixed-ratio objectives are substantially worse than the free low-ratio fits.

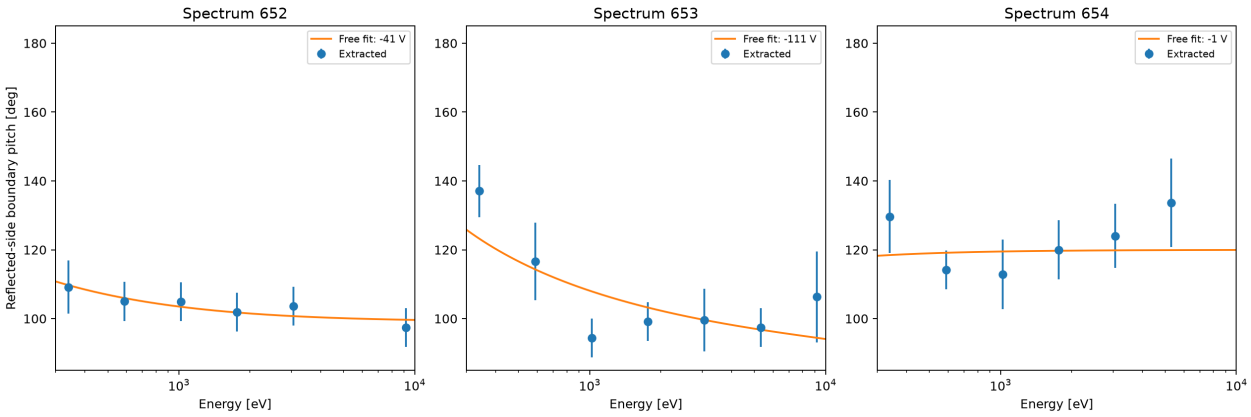


Figure 5: Incident-mean, equal-sector boundary points and preregistered free adiabatic fits. Spectrum 652 is coherent, 653 is non-monotonic, and 654 trends incompatibly with a large negative free potential.

7 Protocol Corrections

All corrections are retained in the experiment history rather than hidden:

1. Phase A2 support integration initially included geometrically available sectors that did not contribute valid observations. The final run uses matched observational support.

2. The first A4 bootstrap pass assigned sequentially different draws to identical row-field and sweep-field inputs. The final pass reuses identical resampling indices.

3. A4 transition nuisance-edge flags and a separate strict-QC analysis were added after provisional QC exposed parameter rails. Preregistered fits remain preserved and primary.

8 Scientific Interpretation

The experiments establish three different levels of evidence.

First, the upward beam is real and robust. Response-aware beam fits repeatedly give a spacecraft-relative negative voltage scale near 145–160 V. This result is stable to normalization order and becomes rail-free when the measured angular width is represented.

Second, the historical joint voltage scale is reproducible. Support integration and fractional energy response move a literal -261 V result into the -151 to -161 V range. This explains how a method of the published form can return approximately -171 V.

Third, the loss-cone boundary does not independently identify both voltage and field ratio on the benchmark distribution. Free boundary fits change with normalization, weighting, nuisance QC, and adjacent spectrum. A publication-like voltage can be obtained from the closest panel by imposing the published magnetic ratio, but the boundary data prefer a substantially lower ratio. The exact published pair is therefore a model-dependent joint solution, not a uniquely selected two-observable measurement in the public reconstruction.

This does not prove that the original authors' private implementation was wrong. It does mean that the paper's qualitative statement that the beam and boundary are mutually consistent cannot be reproduced as a stable independent inference from the archived product and public method description.

9 Consequences for the Project

The robust candidate observable is the beam-centered spacecraft-relative voltage. A mission analysis should treat loss-cone boundaries as conditional diagnostics requiring explicit contrast, width, monotonicity, field-ratio, and parameter-edge gates. Absolute lunar potential relative to ambient plasma requires adding a separately estimated spacecraft potential and its uncertainty.

The negative global spatiotemporal identifiability result is not weakened. A stricter measurement selection can only reduce usable coverage. Quantitative mission statistics must nevertheless be regenerated from a validated observable rather than inherited from the previous joint fit.

10 Final Decision

Experiment 004 does not freeze a production joint loss-cone estimator. It advances the following hierarchy to a one-day validation pilot:

1. response-aware beam-only voltage as the primary candidate observable;
2. direct loss-cone boundary extraction as an identifiability and physical consistency diagnostic;
3. historical joint fit as a provenance comparison only.

The next gate is generalization across every attempted sweep in the hand-vetted 29 April 1999 interval, with complete selection denominators and empirically calibrated beam uncertainty.